Innovation & Entrepreneurship for Engineers

Dr Shankar

IIT Madras

2026

Innovation & Entrepreneurship for Engineers

Module 1
Engineers as Innovators

#1. What is Innovation ?
#2. Innovation through S & T

Module 2
Problem Discovery

#3. Finding Problems worth solving
#4. Customer Discovery & Design Thinking

Module 3
Creativity & Innovation

#5.Innovative Thinking – TRIZ, FLOW, CREATE
#6. Enhancing Innovation – AI - Collaboration

Module 4
Deep-Tech Entrepreneurship

#7. Deep – Tech Ventures
#8. Exponential & Systems Thinking

Module 5
Building a Startup

#9. Economics of Innovation & Entrepreneurship
#10. Business Models
#11. Intellectual Property

Module 6
Prototyping & Presentation

#12. Rapid Prototyping
#13. Pitch Your Idea
#14. Demo Day

Innovation & Entrepreneurship for Engineers

Exponential & Systems Thinking

Dr Shankar

IIT Madras

16 October, 2026

Session Plan

- Exponential Thinking
- Exponential Technology Laws
- Peter Diamandis – 6 D Framework
- Case Studies
- Systems Thinking
- Examples & Case Studies
- Assignment

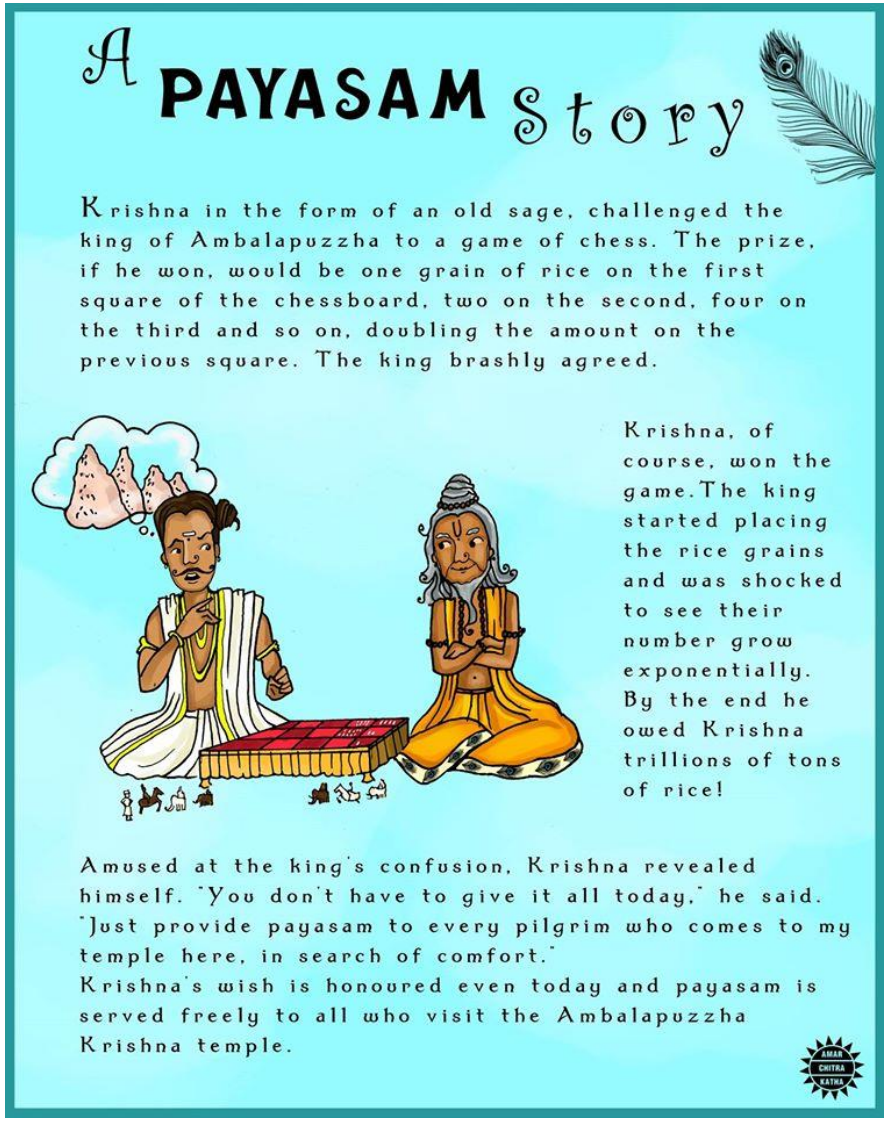
THINK EXPONENTIALLY

From a small innovation idea to a system that can scale

- Exponential Thinking
- Peter Diamandis – 6 D Framework
- Exponential Technology Laws
- Exponential Thinking by Deep Tech Entrepreneurs

Exponential Risce

No	Amount	No	Amount	No	Amount	No	Amount
1	0.01	17	655.36	33	42,949,672.96	49	2,814,749,767,106.56
2	0.02	18	1,310.72	34	85,899,345.92	50	5,629,499,534,213.12
3	0.04	19	2,621.44	35	171,798,691.84	51	11,258,999,068,426.20
4	0.08	20	5,242.88	36	343,597,383.68	52	22,517,998,136,852.50
5	0.16	21	10,485.76	37	687,194,767.36	53	45,035,996,273,705.00
6	0.32	22	20,971.52	38	1,374,389,534.72	54	90,071,992,547,409.90
7	0.64	23	41,943.04	39	2,748,779,069.44	55	180,143,985,094,820.00
8	1.28	24	83,886.08	40	5,497,558,138.88	56	360,287,970,189,640.00
9	2.56	25	167,772.16	41	10,995,116,277.76	57	720,575,940,379,279.00
10	5.12	26	335,544.32	42	21,990,232,555.52	58	1,441,151,880,758,560.00
11	10.24	27	671,088.64	43	43,980,465,111.04	59	2,882,303,761,517,120.00
12	20.48	28	1,342,177.28	44	87,960,930,222.08	60	5,764,607,523,034,230.00
13	40.96	29	2,684,354.56	45	175,921,860,444.16	61	11,529,215,046,068,500.00
14	81.92	30	5,368,709.12	46	351,843,720,888.32	62	23,058,430,092,136,900.00
15	163.84	31	10,737,418.24	47	703,687,441,776.64	63	46,116,860,184,273,900.00
16	327.68	32	21,474,836.48	48	1,407,374,883,553.28	64	92,233,720,368,547,800.00
Total	655.35		42,949,017.60		2,814,706,817,433.60		184,464,625,987,328,000.00



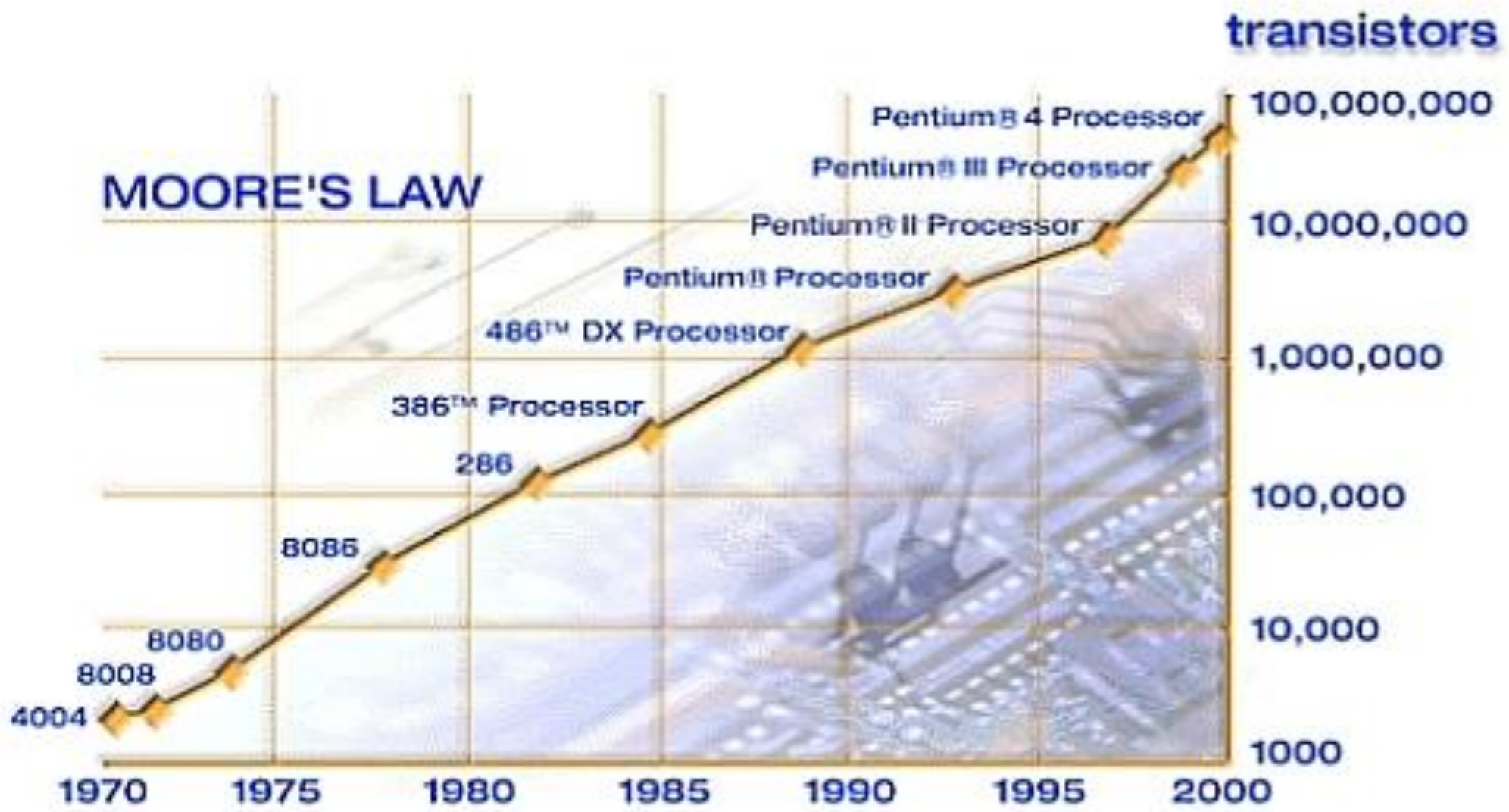
The Second Half of the Chess Board – Ray Kurzweil



$2^{64} - 1 =$
18,446,744,073,709,551,615 grains

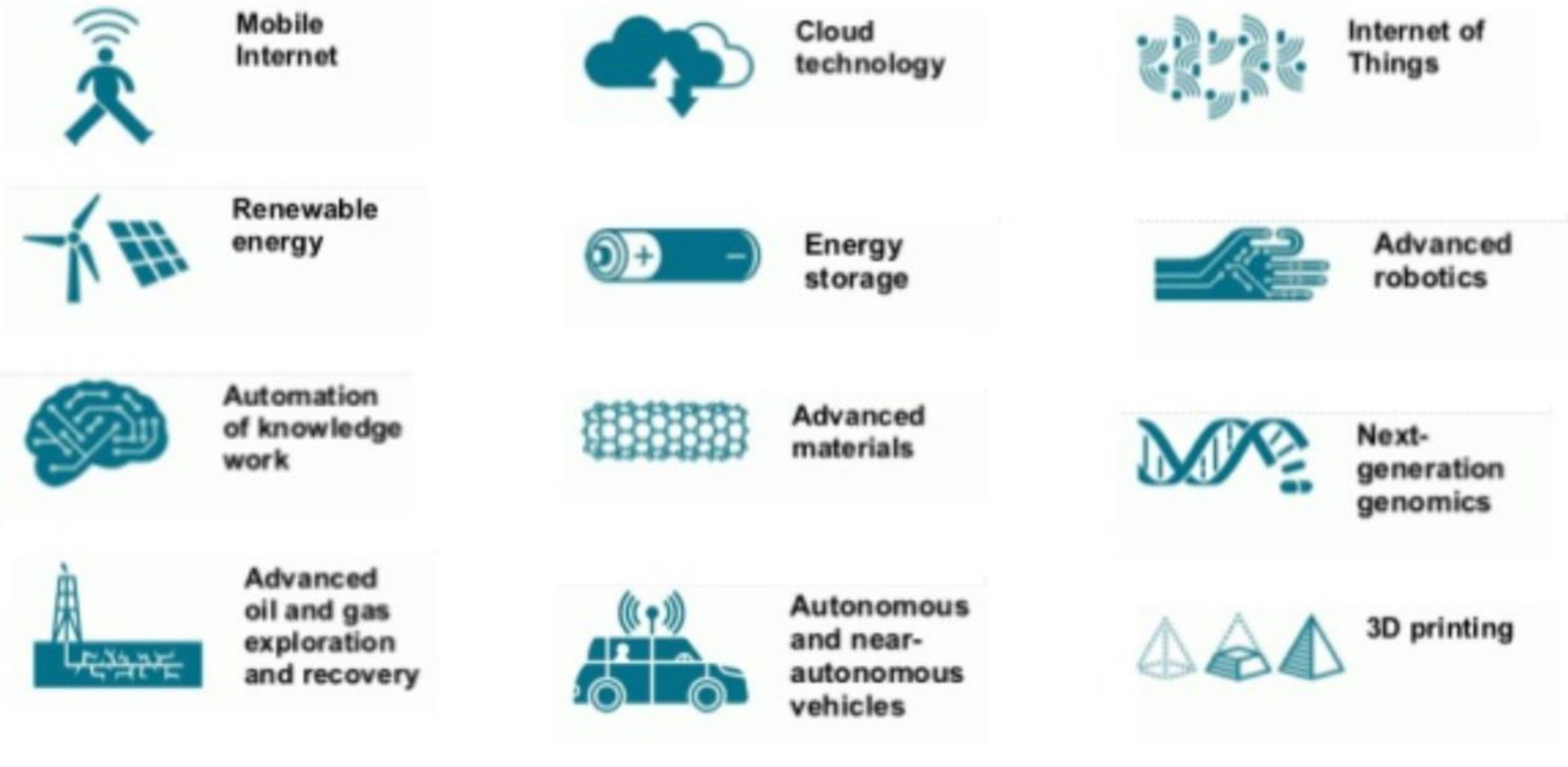
2000 times the global production of rice in 2014 (729,000,000 metric tons)


30 Linear Steps
30 Exponential Steps
30 Meters
25 x


transistors
MOORE'S LAW
Pentium® 4 Processor
Pentium® III Processor
Pentium® II Processor
Pentium® Processor
486™ DX Processor
386™ Processor
286
8086
8080
8008
4004
100,000,000
10,000,000
1,000,000
100,000
10,000
1000
1970
1975
1980
1985
1990
1995
2000

McKinsey Global Institute

Twelve Potentially Economically Disruptive Technologies



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EXPONENTIAL THINKING I VISUAL SUMMARY
Six laws. One converging system.
Each law improves a different part of the technology and business equation.
1 MOORE'S LAW
COMPUTE
Transistor count per chip historically doubled about every 2 years.
More capability in a smaller footprint
2 KRYDER'S LAW
STORAGE
Magnetic storage density grew rapidly; the rate varies by era and technology.
More data retained economically
3 KOOMEY'S LAW
EFFICIENCY
More computations per joule over a historical period; the rate later changed.
More work within an energy budget
4 NIELSEN'S LAW
BANDWIDTH
High-end residential bandwidth grew about 50% per year in a historical series.
5 METCALFE'S LAW
NETWORK EFFECTS
Potential network value is often modelled as proportional to n².
6 WRIGHT'S LAW
FALLING COSTS
Unit costs tend to fall as cumulative production and learning increase.
CONVERGENCE
Capability × Data × Efficiency × Access × Networks × Learning
BUSINESS EFFECT
New products become feasible
Unit economics improve
Network value can grow
New business models emerge
Entrepreneur's question: What customer outcome becomes possible, affordable or scalable when these trends converge?

Technology disruption is a convergence story

A single exponential rarely changes an industry. Several reinforcing curves often do.

COMPUTE

More capability per chip

STORE

More data at lower cost

EFFICIENCY

More work per joule

CONNECT

More bandwidth, lower latency

NETWORK

More useful connections

LEARNING

Lower cost as output scales

CONVERGENCE → NEW CAPABILITIES → NEW BUSINESS MODELS

Moore's Law: more transistors, more possibility

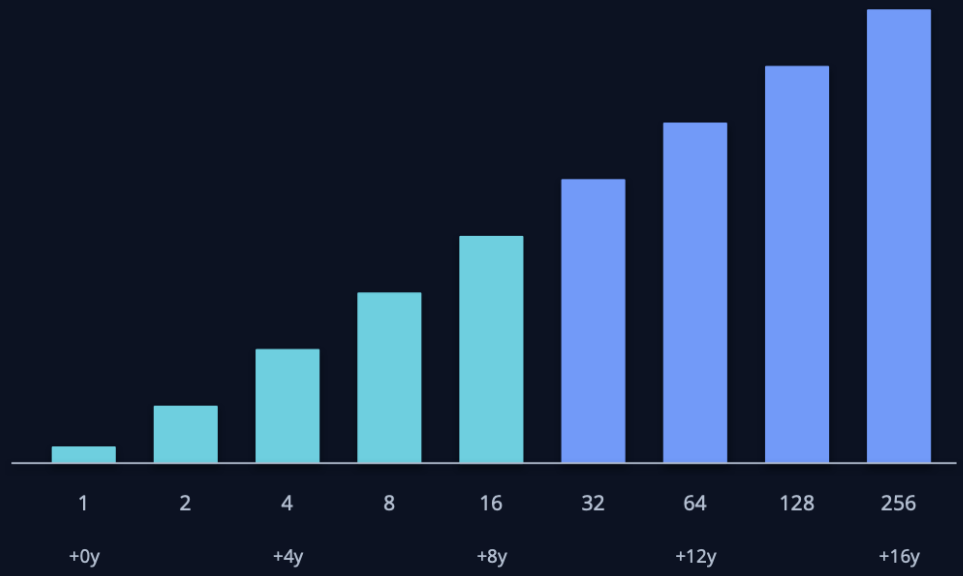
The familiar rule of thumb: transistor count roughly doubles every two years.

2×

transistors per chip in ~2 years (historically)

More integration can enable more compute, memory, control and specialised functions in a similar footprint.

ILLUSTRATIVE INDEX — NOT HISTORICAL CHIP DATA



BUSINESS EFFECT: capabilities once too costly become feasible

Kryder's Law: data becomes economically abundant

Historically, magnetic storage density grew rapidly; the rate has slowed and varies by technology.

STORE MORE

More bits per device and lower cost per bit

Enables archives, media libraries, telemetry histories, model training data and digital twins.



DATA → INSIGHT → SERVICE

Storage alone is not value. Value appears when data can be found, governed, analysed and acted upon.

Koomey's Law: more computation per joule

Historical trend: computation per unit of energy improved exponentially, though the rate changed over time.

COMPUTE / JOULE

Do more work with the same energy

Critical when power, heat, battery life or operating cost is the constraint.



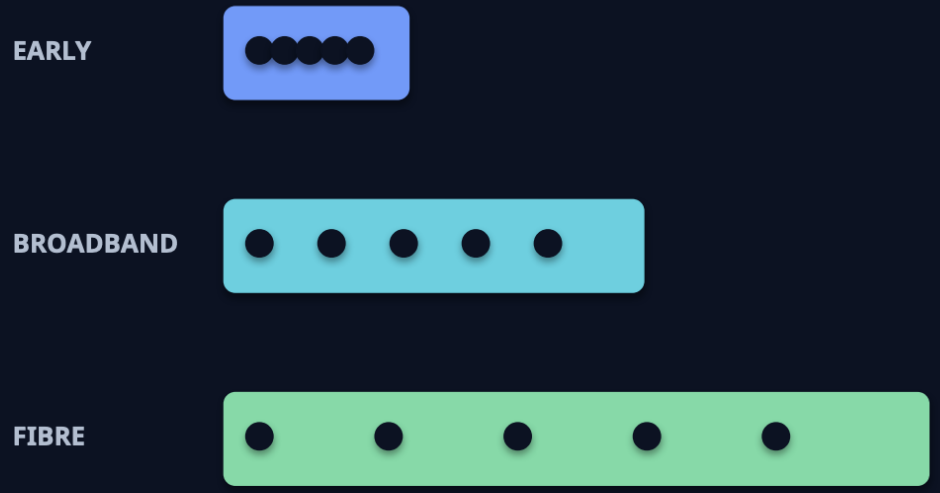
BUSINESS EFFECT

Lower energy per task can unlock edge AI, mobile computing and more economical data-centre services.

Caution: efficiency gains can be offset by rising total demand.

Nielsen's Law: bandwidth expands the feasible experience

A high-end user's internet bandwidth grew by about 50% per year in Nielsen's historical series.



50% / year

≈ 57× in a decade
if the rate compounds

Higher bandwidth supports richer media, cloud applications, remote collaboration and connected products.

BOTTLENECK CHECK: latency • coverage • reliability

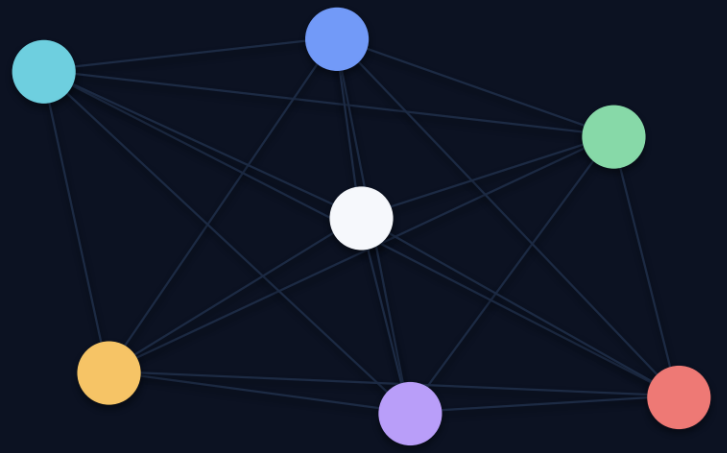
Metcalfe's Law: each new user can make the network more useful

A common heuristic: potential network value grows roughly with the number of possible connections.

$$V \propto n^2$$

More users can create more potential connections.

Examples: marketplaces, professional communities, payment networks, developer ecosystems and communication platforms.



THE CAVEAT

Not every connection is useful. Trust, relevance, quality and congestion matter.

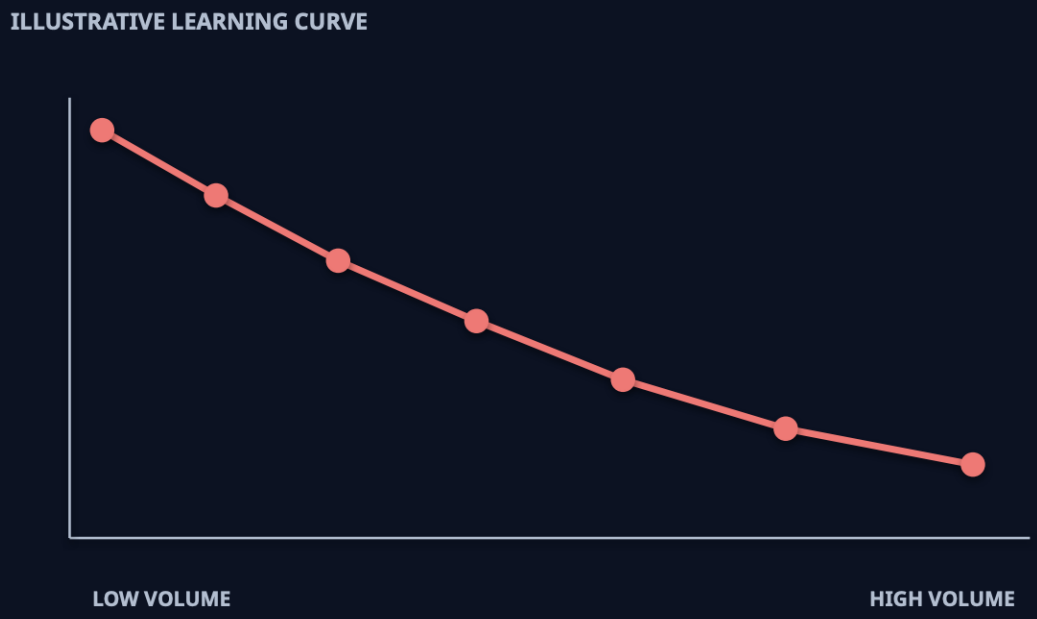
Wright's Law: scale creates learning—and learning lowers cost

Unit cost tends to fall by a recurring percentage as cumulative production doubles.

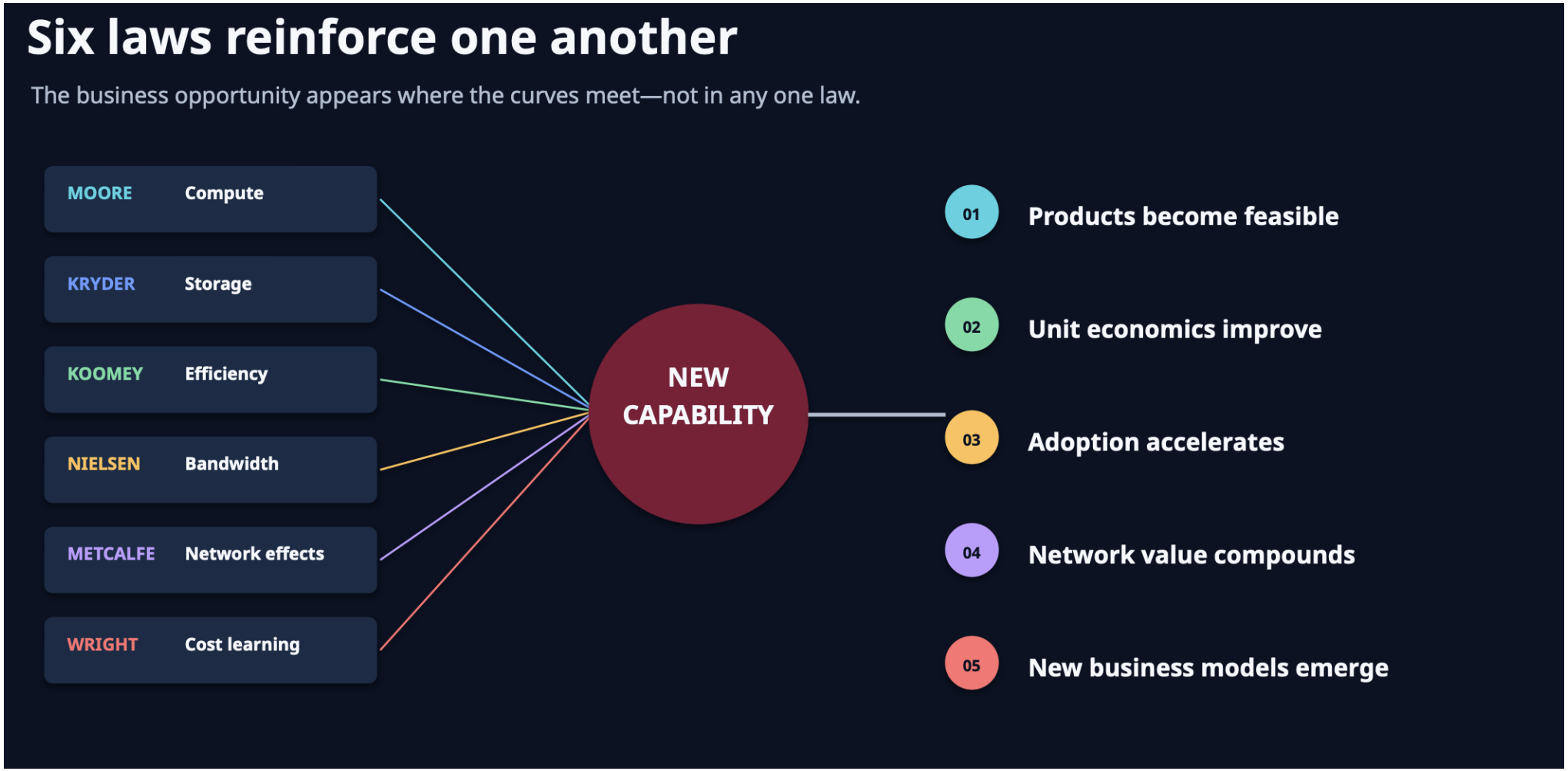
CUMULATIVE OUTPUT ↑

Learning by doing

Better yields • fewer defects • faster assembly • improved design • stronger supplier ecosystem



BUSINESS EFFECT: cost advantage can compound with deployment.


Six laws reinforce one another
The business opportunity appears where the curves meet—not in any one law.
MOORE
Compute
KRYDER
Storage
KOOMEY
Efficiency
NIELSEN
Bandwidth
METCALFE
Network effects
WRIGHT
Cost learning
NEW CAPABILITY
01
Products become feasible
02
Unit economics improve
03
Adoption accelerates
04
Network value compounds
05
New business models emerge

How convergence changes business-model economics

Technology shifts the cost curve; the business model captures the value.

1 INPUT ECONOMICS

- Compute gets cheaper
- Data gets cheaper
- Energy per task falls
- Bandwidth improves

2 PRODUCT DESIGN

- Automate a costly task
- Personalise at scale
- Connect users & assets
- Learn from usage data

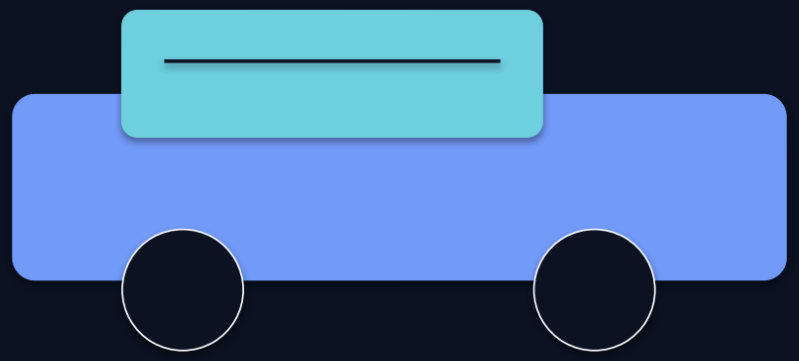
3 VALUE CAPTURE

- Subscription / usage
- Marketplace take rate
- Outcome-based service
- Recurring software revenue

KEY QUESTION: What customer outcome becomes 10× better, cheaper, faster or more accessible?

Example: connected electric mobility

A system-level opportunity emerges when multiple technology curves reinforce each other.



CONNECTED EV

Vehicle + cloud + charging + software

- **MOORE** — On-board compute: perception, control, features
- **KRYDER** — Vehicle data: logs, maps, diagnostics
- **KOOMEY** — Energy-efficient edge processing
- **NIELSEN** — Connected services, OTA and cloud
- **METCALFE** — Charging, fleet and service ecosystems
- **WRIGHT** — Scale and learning lower component costs

Design a venture at the intersection of three laws

10-minute team exercise • choose a real problem, not a technology looking for a problem.

01 PICK A PROBLEM

Whose job is expensive, slow, risky or inaccessible?

02 COMBINE 3 LAWS

Which trends change feasibility or economics?

03 DESIGN THE OFFER

What outcome improves, and for whom?

04 TEST THE LOGIC

What assumption must be validated first?

DELIVERABLE

One sentence: “Because ___ is improving, we can now ___ for ___, and make the model work by ___.”

Exponential trends are powerful—but not infinite

The best innovators know when the curve bends, stalls or hits a constraint.

PHYSICS

Heat, materials, latency and energy

INFRASTRUCTURE

Capex, coverage, supply chains and deployment

ECONOMICS & TRUST

Willingness to pay, privacy, security and regulation

A law is a lens for asking better questions—not a guarantee of tomorrow's outcome.

Metcalfe's Law is a heuristic; Moore, Kryder, Koomey and Nielsen are historical trends; Wright depends on learning and production context.

What is exponential thinking?

LINEAR IMPROVEMENT

Question: How can we make the current product 10% better?

Typical moves:
- improve one feature
- add one sales channel
- optimize a local process

Useful—but may preserve the same underlying system.

EXPONENTIAL POTENTIAL

Question: What changes if a key capability improves rapidly while its cost falls?

Look for:
- reusable software / models
- shared platforms
- network or learning effects
- a redesigned operating model

REALITY CHECK

Not everything scales exponentially.

Constraints may include:
- materials and factories
- energy and compute
- safety and regulation
- trust, service and adoption

A curve needs evidence and a time horizon.

Ask: Which variable could improve by 10×—
and what system change would allow users to benefit from it?

Six laws. One entrepreneurial habit.

Look for the reinforcing system behind a breakthrough.

CAPABILITY × ACCESS × LEARNING × NETWORKS

COMPUTE + STORAGE

Make the previously impossible feasible.

EFFICIENCY + CONNECTIVITY

Make it practical in real-world settings.

NETWORKS + LEARNING

Make it more valuable—and cheaper—as it scales.

FINAL THOUGHT TRIGGER: What is becoming possible now that was impossible—or uneconomic—five years ago?

Moore's Law — Intel
https://www.intel.com/content/www/us/en/newsroom/resources/moores-law.html

Kryder's Law — Scientific American
https://www.scientificamerican.com/article/kryders-law/

Kryder / areal density — Wiley paper
https://doi.org/10.1155/2013/521086

Koomey's Law revisited — Springer
https://link.springer.com/article/10.1007/s10586-024-04767-y

Nielsen's Law — Nielsen Norman Group
https://www.nngroup.com/articles/law-of-bandwidth/

Metcalfe's Law — Information Economics and Policy
https://www.sciencedirect.com/science/article/pii/S0167624516301081

Metcalfe's Law critique — IEEE Spectrum
https://spectrum.ieee.org/metcalfes-law-is-wrong

Wright's Law / learning curves — Our World in Data
https://ourworldindata.org/learning-curve

Learning-by-doing — Journal of Economic Perspectives
https://www.aeaweb.org/articles?id=10.1257/jep.26.3.203

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Peter Diamandis' 6 Ds of Exponentials

1. DIGITIZED

An analog / physical process becomes data or software.

EXAMPLE
Paper maps → digital maps; film photos → digital images.

2. DECEPTIVE

Early gains look tiny because the base is small.

EXAMPLE
Doubling from 0.01 to 0.02 looks unimpressive, despite the doubling.

3. DISRUPTIVE

Capability crosses a threshold where an alternative becomes viable.

EXAMPLE
Digital imaging displaced film for many mainstream uses.

4. DEMATERIALIZED

Several standalone products converge into one device or software service.

EXAMPLE
A smartphone combines camera, GPS, music player and recorder.

5. DEMONETIZED

Price or marginal cost falls sharply; it need not reach zero.

EXAMPLE
Digital distribution can make copying and delivery inexpensive.

6. DEMOCRATIZED

Access expands beyond people previously excluded by cost or expertise.

EXAMPLE
Smartphone tools make computing and information widely accessible.

THE SIX Ds OF EXPONENTIALS

Singularity UNIVERSITY

DIGITIZED

Anything that becomes digitized enters the same exponential growth we see in computing. Digital information is easy to access, share, and distribute. It can spread at the speed of the internet. Once something can be represented in ones and zeros - from music to biotechnology - it becomes an information-based technology and enters exponential growth.





DECEPTIVE

When something starts being digitized, its initial period of growth is deceptive because exponential trends dont seem to grow very fast at first. Doubling .01 only gets you .02, then .04, and so on. Exponential growth really takes off after it breaks the whole-number barrier. 2 quickly becomes 32, which becomes 32,000 before you know it.

DISRUPTIVE

The existing market for a product or service is disrupted by the new market the exponential technology creates because digital technologies outperform in effectiveness and cost. Once you can stream music on your phone, why buy CDs? If you can also snap, store, and share photographs, why buy a camera and film?




DEMONETIZED
Money is increasingly removed from the equation as the technology becomes cheaper, often to the point of being free. Software is less expensive to produce than hardware and copies are virtually free. You can now download any number of apps on your phone to access terabytes of information and enjoy a multitude of services at costs approaching zero.

DEMATERIALIZED

Separate physical products are removed from the equation. Technologies that were once bulky or expensive - radio, camera, GPS, video, phones, maps - are now all in a smartphone that fits in your pocket.





DEMOCRATIZED

Once something is digitzed, more people can have access to it. Powerful technologies are no longer only for governments, large organizations, or the wealthy.

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The 6 Ds are a lens—not a law

Digitization is especially visible in software, data, AI, sensing and digitally controlled processes; physical deep tech often follows a mixed curve.

"Deceptive" means early progress can be hard to notice—not that the technology is misleading.

Disruption occurs when performance, cost, convenience and context cross a meaningful threshold.

Demonetization means a dramatic reduction in cost or price; it does not mean a company cannot make money.

Dematerialization and **Democratization** can be partial: hardware, energy, skills, trust and regulation may remain bottlenecks.

Worked example: AI-enabled crop advisory

DIGITIZED
Field observations, weather, soil and crop images become usable data.

DECEPTIVE
Early models work only for a few crops / districts; measured impact appears small.

DISRUPTIVE
Accuracy and timing become good enough to change a real farming decision.

DEMATERIALIZED
Some expert knowledge becomes available through phone or voice interface.

DEMONETIZED
Shared models and digital delivery may reduce cost per farmer served.

DEMOCRATIZED
Local-language advisory may reach farmers beyond traditional extension capacity.

The 10× Opportunity Canvas

PROBLEM
Who has the problem? How often? What is the cost of doing nothing?

ENABLING CURVE
What capability is improving? What evidence supports the trend?

ADOPTION PATH
Who must use, trust, pay for, regulate or maintain the solution?

FIRST TEST
What two-week experiment could disprove the central assumption?

10× OUTCOME
Which metric could improve by an order of magnitude: cost, time, reach, accuracy, safety or sustainability?

NEW SYSTEM
What can be removed, automated, shared, digitized or redesigned?

BOTTLENECK
Which factor may remain slow or expensive: materials, service, energy, safety or regulation?

Six ventures, one analysis lens

INDIA

UPI ecosystem — interoperable digital payment rails
Ather Energy — electric vehicle + software + charging ecosystem
Sarvam AI — foundation models and Indian-language AI

GLOBAL

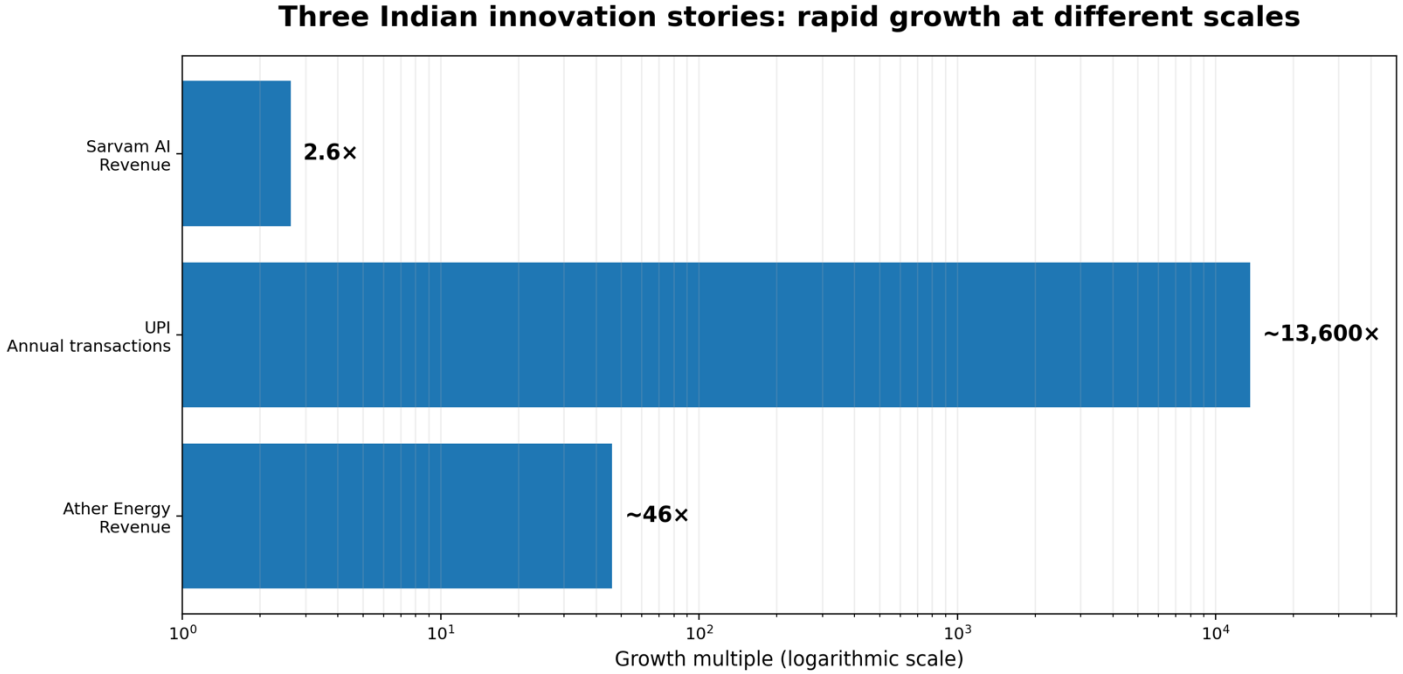
SpaceX — reusability and launch cadence
NVIDIA — GPU + CUDA + developer ecosystem
Amazon Web Services — on-demand computing infrastructure

CASE QUESTIONS

What capability was digitized or made reusable? What looked unimpressive at first? What changed economics, access or adoption? Which constraints did not scale exponentially?

THREE INDIAN INNOVATION STORIES

Rapid growth in digital payments, AI and electric mobility



Different measures and time windows: Sarvam AI and Ather = revenue; UPI = annual transaction volume. Growth multiples show endpoint change, not proof of a constant exponential trajectory.

SARVAM AI | 2.6×
Revenue: ₹11.1 Cr → ₹29.1 Cr (FY24 to FY25)

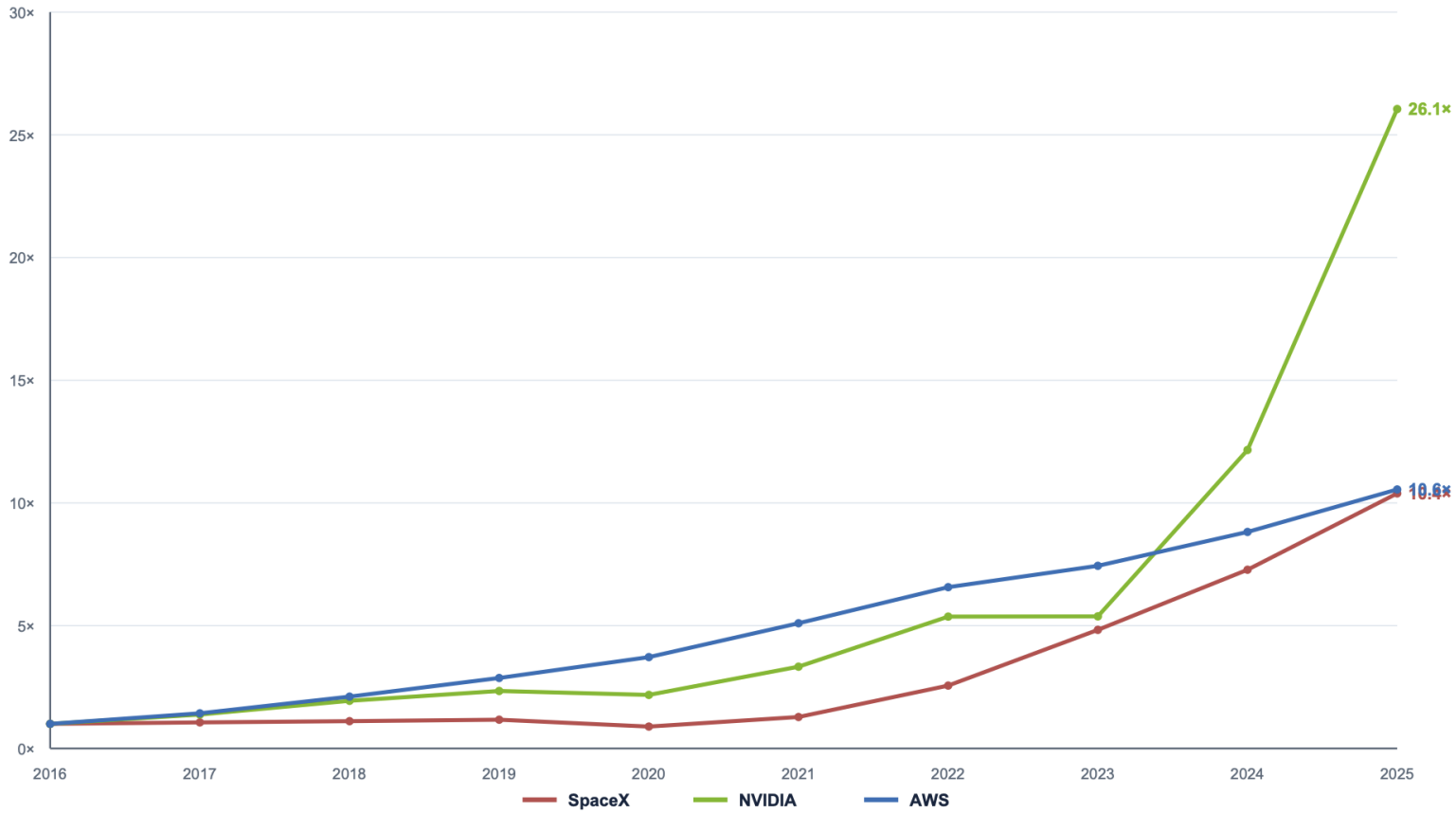
UPI | ~13,600×
Annual transactions: 1.78 Cr → 24,162 Cr (FY16–17 to FY25–26)

ATHER ENERGY | ~46×
Revenue: ₹79.8 Cr → ₹3,672 Cr (FY20 to FY25–26)

TECHNOLOGY BUSINESS GROWTH | 2016–2025

Three technology businesses scaled dramatically

Revenue index: 2016 = 1× for each company • Relative growth, not absolute revenue



Takeaway: NVIDIA's AI-driven revenue surge is especially steep; AWS and SpaceX also scaled substantially.

Caution: SpaceX figures are estimates; NVIDIA uses fiscal years, AWS calendar years. Index values rounded; not proof of a constant exponential rate.

Sources: NVIDIA FY2025 10-K (sec.gov) • Amazon 2025 results (aboutamazon.com) • SpaceX estimates (spacex-stock.com/financials)

CASE 1 • UPI — shared digital rails

ORIGIN STORY

NPCI launched UPI in 2016 in an ecosystem involving banks, apps, merchants and public infrastructure.

CONSTRAINTS

fraud, reliability, digital literacy, dispute handling, connectivity and sustainable economics still require investment.

SCALE MECHANISM

interoperable bank-to-bank payment rails, QR codes and mobile access reduce friction. More users and merchants can increase usefulness across the network.

TEACHING POINT

shared infrastructure can unlock many competing services. Use current NPCI data if quoting transaction volumes.

Which of the 6 Ds best explains the scale mechanism?

Which D is not yet relevant or would be misleading?

CASE 2 • Ather Energy — product + platform

ORIGIN STORY

founded in 2013 by Tarun Mehta and Swapnil Jain; built an integrated electric two-wheeler technology platform.

SCALE MECHANISM

vehicle hardware + software + connected features + Ather Grid charging + a product platform spanning performance and family scooters.

CONSTRAINTS

manufacturing, service, battery cost, charging coverage, supply chain and affordability do not scale automatically.

TEACHING POINT

the product is not just the scooter; it includes charging, software, service, manufacturing and customer trust.

Which of the 6 Ds best explains the scale mechanism?

Which D is not yet relevant or would be misleading?

CASE 3 • Sarvam AI — AI for Indian languages

ORIGIN STORY

founded in 2023 by Vivek Raghavan and Pratyush Kumar; focused on full-stack AI for Indian-language needs.

SCALE MECHANISM

reusable foundation models can support speech, translation, text, vision and agent workflows; deployment tools can broaden access.

CONSTRAINTS

compute cost, data quality, language diversity, reliability, enterprise adoption and global competition.

TEACHING POINT

a young venture's strategic intent is not proof of success. Measure language quality, real usage and unit economics.

Which of the 6 Ds best explains the scale mechanism?

Which D is not yet relevant or would be misleading?

CASE 4 • SpaceX — reuse as system redesign

ORIGIN STORY

founded in 2002 to lower the cost and increase the reliability of access to space.

SCALE MECHANISM

reusable boosters, high flight cadence, vertical integration and rapid iteration seek to spread fixed costs across more launches.

CONSTRAINTS

reuse requires inspection, refurbishment, reliability, launch operations, regulation and sufficient demand.

TEACHING POINT

changing unit economics can require redesigning operations and feedback loops—not just components.

Which of the 6 Ds best explains the scale mechanism?

Which D is not yet relevant or would be misleading?

CASE 5 • NVIDIA — platform leverage

ORIGIN STORY

founded in 1993; GPU architecture began in graphics, while CUDA made GPUs programmable for broader parallel computing.

SCALE MECHANISM

hardware + developer tools + libraries + systems + ecosystem. Improvements across layers reinforce adoption.

CONSTRAINTS

semiconductor supply, power, cost, competition, software portability and customer dependence remain real.

TEACHING POINT

an ecosystem can compound value. What could your project make easier for thousands of other builders?

Which of the 6 Ds best explains the scale mechanism?

Which D is not yet relevant or would be misleading?

CASE 6 • AWS — capability on demand

ORIGIN STORY

Amazon faced the cost and delay of provisioning IT infrastructure; AWS launched S3 and EC2 in 2006.

SCALE MECHANISM

on-demand, pay-as-you-go infrastructure reduces upfront capital and time-to-experiment for developers and companies.

CONSTRAINTS

cloud costs can rise; security, lock-in, reliability, latency and skills affect real-world use.

TEACHING POINT

convert a scarce capability into a service with low entry cost and fast access.

Which of the 6 Ds best explains the scale mechanism?

Which D is not yet relevant or would be misleading?

What the six cases have in common

UPI: interoperable shared rails can create network effects across banks, apps, merchants and users.

Ather: a physical product gains leverage from software, charging, service and a product platform.

Sarvam: reusable models can support multiple products, but quality and unit economics must be proven.

SpaceX: reuse and cadence target a different cost curve through system-level redesign.

NVIDIA: hardware/software ecosystem and developer adoption reinforce each other.

AWS: infrastructure-as-a-service lowers the cost and time required to experiment and scale.

EXPAND checklist

Use this to evaluate whether your exponential hypothesis has a mechanism.

E — ENABLING CURVE
What capability is improving? What evidence / trend supports it?

X — 10× OUTCOME
Which measurable outcome could improve 10×, and by when?

P — PLATFORM
Can one core capability serve multiple users, products or contexts?

A — ACCESS
Can cost, skills or infrastructure barriers to adoption fall?

N — NETWORK / LEARNING
Does use create data, learning, trust, partners or network effects?

D — DEPENDENCIES
What bottleneck could stop scale: hardware, energy, safety, regulation or supply?

Team worksheet • Exponential Project Canvas

PROJECT / USER
Target user: ____ Painful problem: ____

CURRENT BASELINE
Today's cost / time / reach / accuracy / safety: ____

10× ASPIRATION
Improve ____ by 10× within ____; baseline evidence: ____

ENABLING TECHNOLOGY
Capability curve: ____ Evidence/source: ____

6 DS
Relevant Ds and why: ____ Not-yet-relevant Ds: ____

SYSTEM REDESIGN
What can be digitized, automated, shared, removed or redesigned? ____

BOTTLENECK / RISK
Biggest non-exponential constraint + unintended risk: ____

2-WEEK EXPERIMENT
Test ____ with ____; success threshold + failure criterion: ____

Team assignment • Turn your project into an exponential hypothesis

1. Describe your current solution and baseline in one sentence plus one measurable metric.

2. Propose a 10× target: cost, time, reach, accuracy, safety, convenience or sustainability.

3. Apply the 6 Ds: explain at least two relevant Ds and the causal mechanism.

4. Draw the system: user → product/service → infrastructure/partners → value/revenue.

5. Identify one non-exponential bottleneck and one ethical, inclusion or environmental risk.

6. Design a two-week experiment that could prove your hypothesis wrong.

One-minute individual reflection

My project’s most promising 10× metric is…

The technology / system shift that could enable it is…

The 6 D(s) most relevant to our idea are… because…

The assumption I most need to test is…

The experiment I will run next is…

Session Plan

- Exponential Thinking
- Exponential Technology Laws
- Peter Diamandis – 6 D Framework
- Case Studies
- Systems Thinking
- Examples & Case Studies
- Assignment